SNOWFLAKE Workshop

Bank Mandiri

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Day 1 - Session 1

- **Session 1: Building RAG Applications with Snowflake Cortex AI & Snowflake Intelligence**
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  - Introduction to Snowflake Cortex AI
    - Overview of Cortex AI ecosystem: LLM Functions (AI_COMPLETE, AI_EMBED, AI_EXTRACT, AI_SUMMARIZE), Cortex Search, Cortex Analyst, and Cortex Agents. Understanding the unified AI platform.
  - Processing Unstructured Data
    - Ingesting documents (PDFs, text files) into Snowflake internal stages. Using `AI_PARSE_DOCUMENT` for text extraction. Chunking strategies with `SPLIT_TEXT_RECURSIVE_CHARACTER`. Generating vector embeddings with `AI_EMBED`.
  - Building a Cortex Search Service
    - Creating a Cortex Search Service for hybrid search (semantic + keyword). Configuring source queries, target lag, and embedding models. Querying the search service via SQL and Python.
  - RAG on Structured Data with Cortex Analyst
    - Creating Semantic Views/Models for structured table querying. Enabling natural language to SQL via Cortex Analyst. Best practices for semantic model design (metrics, dimensions, time grains).
  - Building a Cortex Agent
    - Combining Cortex Search (unstructured) and Cortex Analyst (structured) into a single Cortex Agent. Configuring multi-tool orchestration. Testing the agent with natural language queries across both data sources.
  - Deploying with Snowflake Intelligence
    - Publishing the agent as a Snowflake Intelligence application. Sharing the AI assistant with business users. Monitoring agent usage and response quality via AI Observability.

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Day 1 - Session 2

- **Session 2: Data Protection Policies in Snowflake**
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    - Snowflake Access Control Fundamentals
        - Overview of Snowflake's security architecture. Role-Based Access Control (RBAC): roles, privileges, role hierarchy. Discretionary Access Control (DAC). Best practices for banking environments: separation of duties, least privilege principle.
    - Data Masking
        - Creating masking policies for sensitive columns (e.g., account numbers, customer names, KTP/NIK). Conditional masking based on roles. Tag-based masking policies for scalable governance. Practical example: masking PII for analyst roles while allowing full access for compliance roles.
    - Row Access Policies
        - Creating row access policies to restrict data visibility per role/user. Using mapping tables for flexible access control. Real-world scenario: branch-level data segregation — each branch sees only its own customer records. Interaction with data sharing and secure views.
    - Projection Policies
        - Understanding projection policies: preventing SELECT on specific columns without WHERE clause restrictions. Use case: preventing bulk extraction of sensitive data while allowing filtered queries. Combining projection policies with masking and row access policies.
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Day 2 - Session 3

- **Session 3 : End-to-End Machine Learning on Snowflake**
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  - **Snowflake ML Platform Overview**
    - Introduction to Snowflake ML: unified platform for end-to-end ML. Key components: Notebooks on Container Runtime, Feature Store, Model Registry, ML Jobs, ML Observability, ML Lineage. Why Snowflake for ML: data proximity, governance, no data movement.
  - **Data Preparation & Exploration**
    - Using Snowflake Notebooks (Jupyter-compatible) for data exploration. Data loading and transformation with Snowpark Python DataFrames. Exploratory data analysis with built-in visualization. Handling structured data from Snowflake tables.
  - **Feature Engineering & Feature Store**
    - Creating features using Snowflake ML APIs and Snowpark transformations. Registering features in the Snowflake Feature Store. Managed vs. user-managed feature pipelines. Automated feature refresh from batch and streaming sources. Feature versioning and discovery.
  - **Model Training**
    - Training models using popular frameworks (scikit-learn, XGBoost, LightGBM) in Snowflake Notebooks.
  - **Model Deployment, Registry & Inference**
    - Logging trained models into the Snowflake Model Registry. Model versioning, tagging, and aliasing. Batch inference using SQL and Python. Online/real-time inference via REST API endpoints. Deploying models to Snowpark Container Services (SPCS).
  - **Model Operations, Monitoring & ML Lineage**
    - ML Observability: monitoring model performance, drift detection, and alerting. Model explainability with SHAP values. ML Lineage: tracing end-to-end dependencies from source data → features → datasets → models. Ensuring reproducibility and compliance for banking regulatory requirements.

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Day 2 - Session 4

- **Session 4 : Accelerating ML Development with Cortex Code**
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  - **Introduction to Cortex Code**
    - What is Cortex Code: AI-driven coding agent integrated into Snowflake. Two interfaces: Cortex Code in Snowsight (web UI) and Cortex Code CLI (terminal). Key capabilities: code generation, SQL optimization, data analysis, agentic workflows. Deep Snowflake awareness: RBAC, schemas, SQL syntax, best practices.
  - **Cortex Code in Snowsight**
    - Accessing Cortex Code from the Snowsight assistant panel. Natural language to SQL generation. Code explanation and refactoring. Quick actions: fix errors, format code, explain logic. Context-aware assistance within Workspaces and Notebooks.
  - **Building ML Models with Cortex Code**
    - Live demo: asking Cortex Code to build a complete ML model from scratch. Generating synthetic data for testing. Creating Feature Store definitions. Training and evaluating models — all via natural language prompts.

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Thank You

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